Batterjee Medical College
For Science and Technology

Psychiatry · Clinical Lecture

Substance Use Disorders

From the neurobiology of addiction to the substances of abuse — diagnosis, intoxication and withdrawal, and evidence-based management. DSM-5-TR aligned, with a Saudi and regional focus.

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Learning Objectives (1 of 2)

Mapped to the BMC Knowledge / Skill / Value framework. Prerequisite: psychiatric history taking and the mental status examination.

Knowledge & understanding

- Explain addiction neurobiologically and biopsychosocially
- Separate tolerance, physical dependence and addiction
- State the DSM-5 criteria and the severity grading

Skills

- Take a substance history one substance at a time
- Screen with CAGE or AUDIT and deliver a brief intervention
- Interpret a urine drug screen — including what it misses

Values & attitudes

- Treat addiction as a chronic relapsing illness, not a moral failing
- Read relapse as a signal to adjust treatment, not as failure
- Ask about substance use without judgement

Learning Objectives (2 of 2)

Knowledge & understanding

- Name the intoxication and withdrawal picture of each major class
- Know which withdrawal syndromes are life-threatening
- Outline medication-assisted and psychosocial treatment

Skills

- Recognise delirium tremens, opioid overdose and Wernicke's, and act
- Match your approach to the patient's stage of change
- Treat comorbid illness concurrently, not after abstinence

Values & attitudes

- Mobilise faith, family and community as protective factors
- Do not withhold analgesia from a dependent patient
- Address stigma — the main barrier to presenting for treatment

The two objectives that examiners and wards test hardest are diagnosing SUD with DSM-5 and recognising the intoxication and withdrawal syndromes — because those drive immediate management.

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PART 1

Foundations of Addiction

What addiction is · what it does to the brain · how it is diagnosed · what drives it · and its consequences.

Substance use disorder & addiction

Substance use disorder

- Recurrent use of alcohol or other drugs causing clinically significant impairment
- Health problems, disability, and failure to meet major responsibilities at work, school or home
- Graded mild / moderate / severe by the number of criteria met

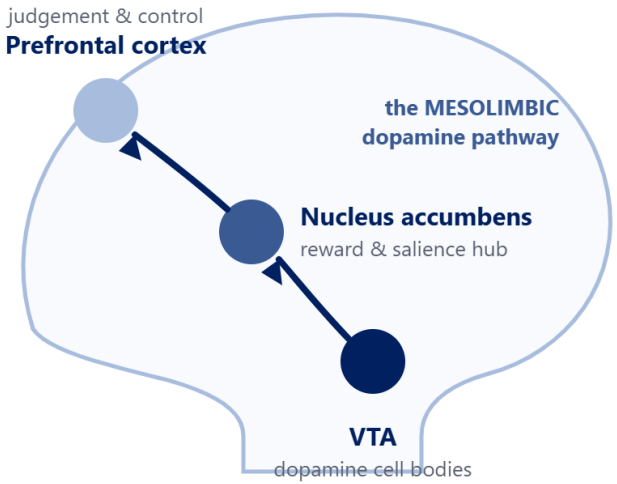
Addiction

- The most severe, chronic stage
- Substantial loss of self-control, with compulsive use despite a desire to stop
- In DSM-5, "addiction" corresponds to severe substance use disorder

Severity by criteria count: 2–3 mild · 4–5 moderate · ≥ 6 severe. The word "addiction" is reserved for the severe end of a continuum.

Addiction is a brain disease

All addictive drugs converge on one final common pathway — the mesolimbic dopamine reward circuit.



What every addictive drug does

Causes a dopamine surge far larger than any natural reward

The brain "learns" the drug is supremely important

Conditioning stronger than food, relationships or safety

Addiction is a brain disease, not a moral failing

All addictive drugs converge on this one final common pathway — the same circuit antipsychotics act on in psychosis.

Addiction is a brain disease, not a moral failing. The same figure used for antipsychotics in the Psychopharmacology lecture explains addiction here — the mesolimbic pathway is the reward circuit every addictive drug hijacks.

What happens to the brain — tolerance & craving

This neuroadaptation explains why willpower alone fails: the reward system itself has been re-set.

↘ Down-regulation

Repeated surges → the brain makes less dopamine and fewer receptors.

Anhedonia & tolerance

Natural pleasures feel flat; larger doses are needed for the same effect.

Cue reactivity

Receptors fire to cues, not just the drug → craving. Use shifts from getting "high" to feeling "normal".

Tolerance vs. dependence vs. addiction

Term	What it means	Is it a disease?
Tolerance	Needing more for the same effect	No — a normal physiological adaptation
Physical dependence	Withdrawal on stopping	No — also normal with many drugs (chronic opioids, β-blockers)
Psychological dependence	Craving and compulsion to use	Part of the syndrome
Addiction (severe SUD)	Compulsive use despite harm, plus loss of control	YES — this is the disease itself

A cancer patient on long-term opioids has tolerance and physical dependence but is NOT addicted. Addiction is the behavioural syndrome, not the physiology alone.

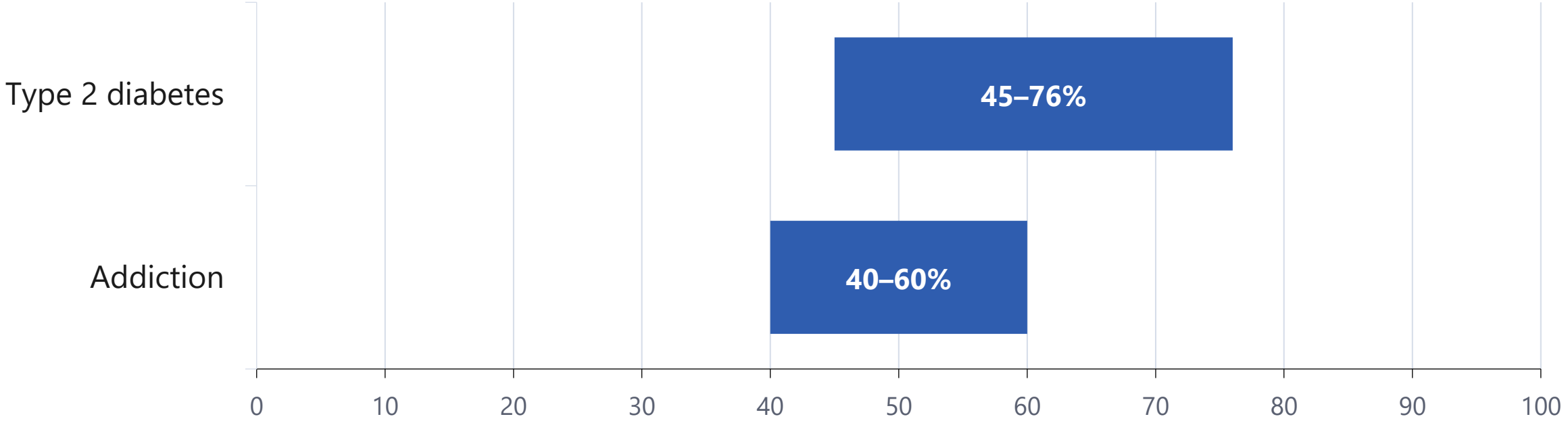
Addiction behaves like other chronic diseases

40–60%
heritability

40–60%
relapse after treatment

Relapse ≠ failure
it signals a need to adjust treatment

Relapse rates after treatment (%) — addiction compared with type 2 diabetes



Persistent physiological change; abstinence and medication help but do not simply "cure". Relapse after treatment is the norm in chronic disease — it is not a failure of willpower.

DSM-5 criteria — ≥ 2 of 11 in 12 months

The eleven criteria cluster into four intuitive groups. Severity: 2–3 mild · 4–5 moderate · ≥ 6 severe.

Impaired control

Larger or longer than intended · unsuccessful attempts to cut down · much time getting, using or recovering · craving.

Social impairment

Role failure at work, school or home · interpersonal problems · activities given up.

Risky use

Use in hazardous situations · use despite physical or psychological harm.

Pharmacological

Tolerance · withdrawal.

Addiction — the "4 C's"

4 C's

A quick clinical gestalt that maps onto the DSM-5 domains — useful when you are taking a history and do not have the full checklist to hand.

C Compulsion

A compulsion to use

C Craving

Persistent desire for the substance

C Control

Loss of control over amount and frequency

C Consequences

Continued use despite consequences

What increases the risk of addiction?

Susceptibility is multifactorial — genes, environment and development interact to shape who initiates, continues, and progresses.

Genetic / biological

Heritability, family history, and modelling within the family.

Early exposure

Adolescence — the developing brain is the most vulnerable period.

Environment

Poor support, easy access, normative attitudes, high-risk settings.

Mental illness

Mood disorders, anxiety, PTSD, ADHD.

Consequences of addiction

Health

- Organ damage
- Overdose; coma and death
- Suicide

Psychosocial

- Accidents; relationship breakdown
- Child neglect and abuse
- Unemployment, poverty, homelessness
- Legal problems

Spiritual

- Disconnection from faith — God and Islam
- Faith and community are also a powerful PROTECTIVE factor
- Worth mobilising deliberately in treatment

Health consequences by substance

Substance	Major health harms
Alcohol	Liver disease (fatty liver → hepatitis → cirrhosis), cognitive impairment, ataxia and balance problems
Tobacco	Lung cancer; head and neck cancers; COPD; cardiovascular disease
Cannabis	Memory deficits; link to psychosis and schizophrenia — adolescents particularly at risk
Cocaine	Arrhythmia, myocardial infarction, stroke
Opioids	Respiratory depression; a recent sharp rise in overdose deaths

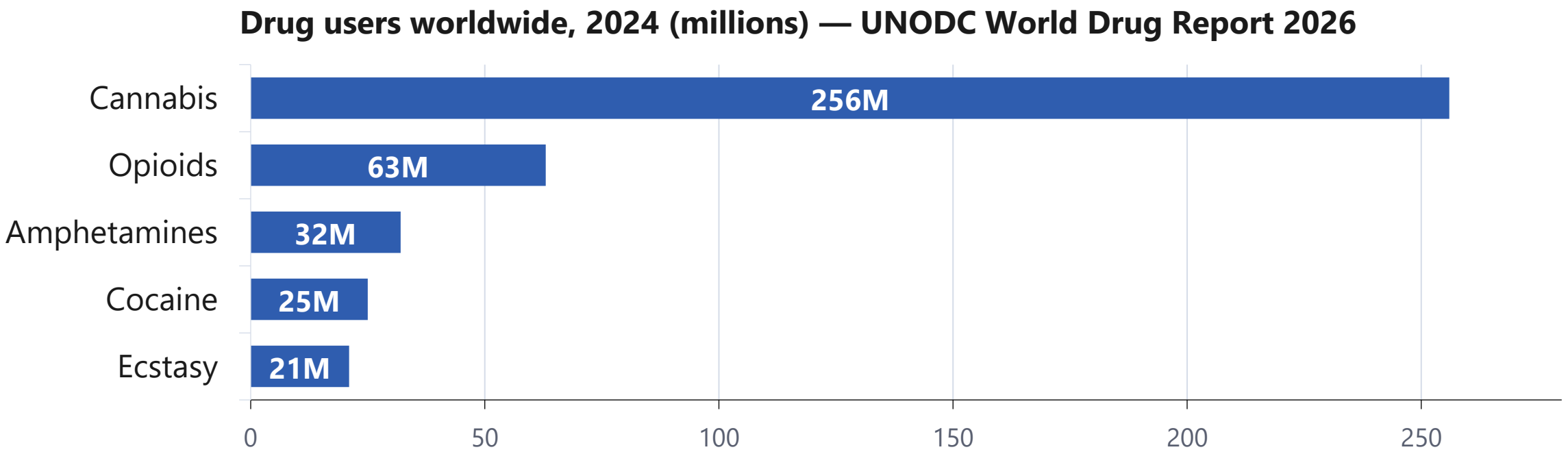
Each class has a signature organ toxicity worth memorising — these are the harms that motivate treatment and that patients most need to hear.

The global scale of substance use

~8M
annual deaths — tobacco

2.6M
annual deaths — alcohol

0.6M
annual deaths — all illicit drugs



The key teaching point: the LEGAL drugs — tobacco (~8M deaths/yr) and alcohol (2.6M) — cause far more death than all illicit drugs combined (~0.6M), while cannabis is by far the most widely used. Weigh harm by burden, not by legal status. Sources: UNODC World Drug Report 2026 (2024 data); WHO 2024.

The major classes of substances

Group them by what they do to the CNS — the grouping predicts the clinical syndromes.

Depressants — switch it off

- Alcohol
- Sedatives / hypnotics (benzodiazepines)
- Opioids
- Inhalants

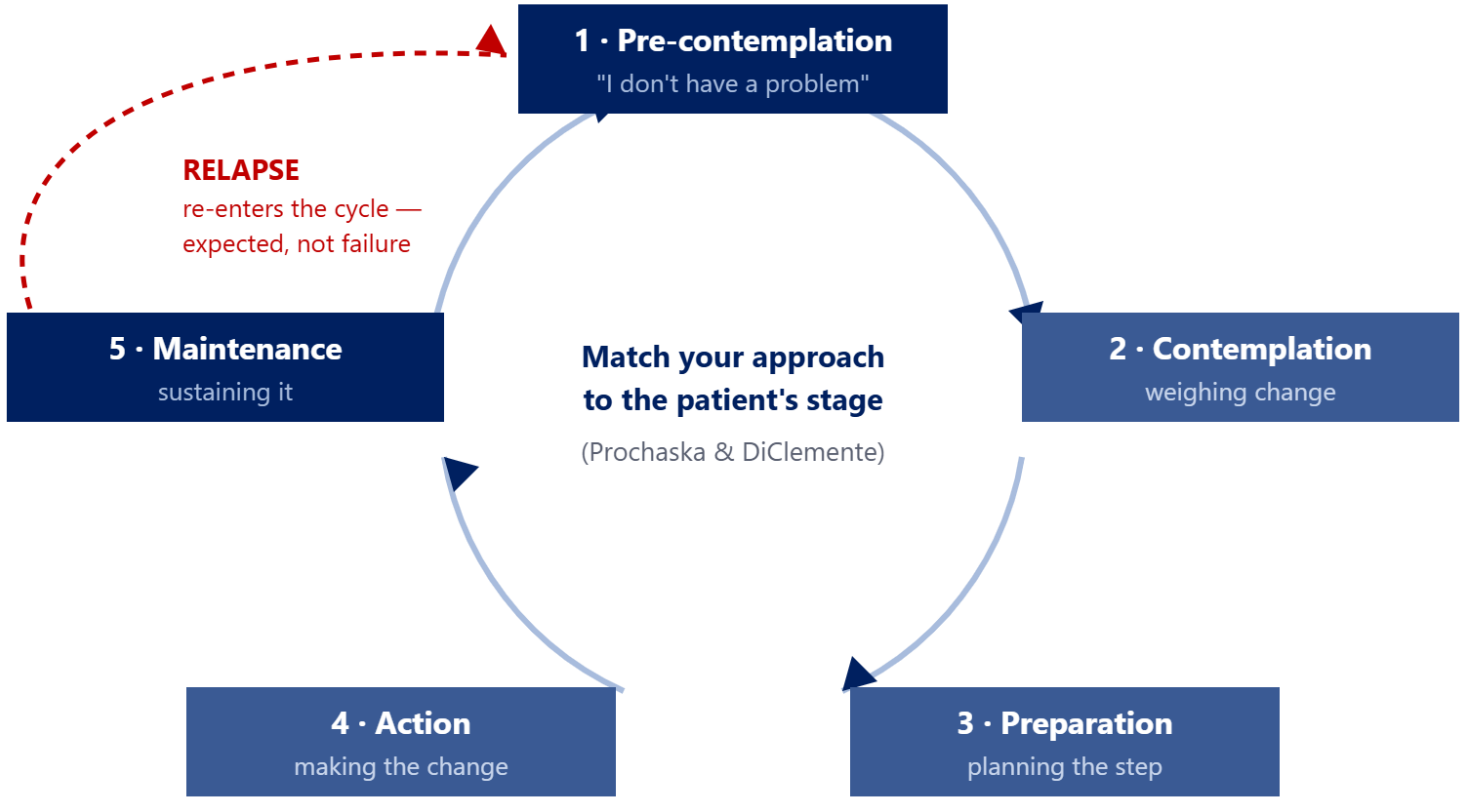
Stimulants — switch it on

- Cocaine
- Amphetamine / methamphetamine
- Captagon, khat
- Caffeine, nicotine

Hallucinogens · cannabis · NPS

- LSD, psilocybin
- Cannabis
- Novel psychoactive substances

Screening & brief intervention



Motivational interviewing: a collaborative, non-confrontational style — roll with resistance, develop discrepancy, and support the patient's own reasons and autonomy to change.

Interpreting the urine drug screen

Approximate detection windows

- Alcohol ~12 h (EtG longer)
- Amphetamines, cocaine, opioids: 1–3 days
- Benzodiazepines: days to weeks (agent-dependent)
- Cannabis: up to ~30 days in chronic users

The traps

- False positives: pseudoephedrine / bupropion → amphetamine; dextromethorphan / ketamine → PCP; poppy seeds → opiates
- Standard screens MISS: fentanyl and synthetic opioids, methadone / oxycodone, many benzodiazepines (clonazepam, lorazepam), synthetic cannabinoids and cathinones

A screen is presumptive, not diagnostic — confirm important results with GC-MS/LC-MS. A "negative opiate screen" does not exclude fentanyl.

Psychotherapeutic interventions in addiction

Psychosocial treatment is the backbone across every substance — medication supports it, not the other way round.

Individual

Motivational enhancement · CBT and relapse prevention (triggers, high-risk situations, coping skills) · contingency management, rewarding verified abstinence.

Group & mutual help

12-step facilitation · AA and NA — fellowship and sponsorship · group therapy · therapeutic communities.

Family & education

Family or couples therapy to reduce enabling and rebuild support. Education underpins all of it.

Dual diagnosis

The picture

- Co-occurring SUD plus a psychiatric disorder — very common
- Mood, anxiety, PTSD, ADHD, psychosis, personality disorders
- Bidirectional: substances cause, mimic or worsen symptoms; illness drives "self-medication"

Why it matters

- Worse prognosis; raised suicide risk, non-adherence, relapse
- Treat both concurrently — integrated care
- Do NOT wait for abstinence before treating the mental illness

Always re-assess psychiatric symptoms after a period of abstinence — many substance-induced states resolve, clarifying the true diagnosis.

Gambling disorder & substances in pregnancy

Gambling disorder

- The only behavioural addiction in DSM-5
- Engages the same reward circuitry as drugs
- Tolerance-like (bigger bets), withdrawal-like (restless when stopping), and chasing losses

Substances in pregnancy

- Alcohol → Fetal Alcohol Syndrome — a leading preventable cause of intellectual disability
- Opioids → Neonatal Abstinence Syndrome
- Tobacco → low birth weight, prematurity
- Cocaine → placental abruption

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PART 2

Depressants

Alcohol · sedatives and hypnotics · opioids · inhalants — the CNS depressants. Watch which withdrawals can kill.

Alcohol intoxication — DSM-5 criteria

A · B · D

- A. Recent ingestion of alcohol
- B. Clinically significant problematic behavioural or psychological change — inappropriate sexual or aggressive behaviour, mood lability, impaired judgment
- D. Not attributable to another medical condition or better explained by another mental disorder

C · ≥ 1 sign during or shortly after

- Slurred speech
- Incoordination
- Unsteady gait
- Nystagmus
- Impairment in attention or memory
- Stupor or coma

A CNS depressant — at high blood levels it kills by respiratory depression, especially combined with other depressants.

Blood alcohol level & CNS effects

Blood alcohol (mg%)	Effect
20–50	Relaxation, mild euphoria, lowered inhibitions
~80 (0.08%)	Impaired coordination and judgment — the common legal driving limit where alcohol is permitted
100–200	Slurred speech, ataxia, incoordination, mood lability
200–300	Nausea and vomiting, marked confusion, stupor
> 400	Coma, respiratory depression, death

Tolerance shifts the curve: a chronic drinker may look sober at a level that would render a naïve drinker comatose — never judge intoxication by appearance alone.

Screening & laboratory markers

CAGE questionnaire (≥ 2 = concern)

- C — have you tried to Cut down?
- A — Annoyed by criticism of your drinking?
- G — felt Guilty about drinking?
- E — Eye-opener drink in the morning?

Labs of heavy drinking

- Raised GGT (> 35 IU/L)
- Raised MCV (macrocytosis)
- AST:ALT > 2:1
- Blood alcohol level

None of these diagnoses a disorder on its own — they raise suspicion and open the conversation.

Risk factors for alcohol use disorder

Personality

Impulsive, gregarious, less conforming — or isolated and avoidant.

Occupation

High-stress roles with easy access, irregular hours, or a culture that discourages help-seeking.

Psychosocial stress

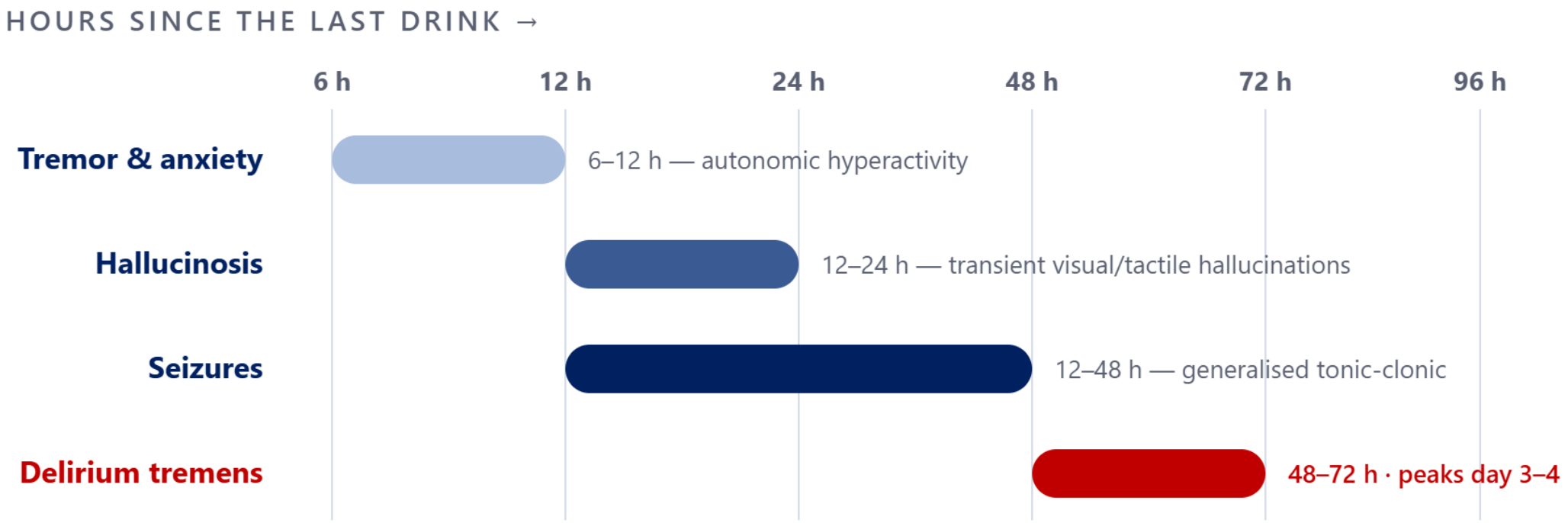
Social isolation; financial, occupational or academic difficulty; marital conflict.

Emotional

Anxiety, chronic insomnia, depression.

WHO: alcohol kills more people worldwide than AIDS, tuberculosis or violence — a legal, socially embedded drug is among the most harmful of all.

Alcohol withdrawal — DSM-5 criteria



Alcohol withdrawal can be fatal — benzodiazepines + thiamine + fluids; DT mortality ~1–5% treated, ~15–35% untreated.

Unlike opioid or stimulant withdrawal, ALCOHOL WITHDRAWAL CAN BE FATAL — seizures and delirium tremens. Learn the clock.

Delirium tremens (DTs)

The picture

- Severe withdrawal; onset 48–72 h, peaks day 3–4, lasts 3–5 days
- Delirium + gross tremor + autonomic storm
- Dehydration and electrolyte disturbance; insomnia
- Mortality ~1–5% treated, up to ~15–35% untreated

Management

- A medical emergency — ICU or medical ward
- Benzodiazepines; thiamine; fluids and electrolytes
- Guard against seizures; adjust surroundings
- AVOID antipsychotics as monotherapy — they lower the seizure threshold

Wernicke–Korsakoff

	Wernicke's encephalopathy	Korsakoff syndrome
Nature	Acute neurological EMERGENCY	Chronic, often irreversible
Cause	Thiamine (B1) deficiency	Untreated or repeated Wernicke's
Features	Triad: confusion · ophthalmoplegia · gait ataxia	Anterograde amnesia + confabulation
Action	GIVE THIAMINE BEFORE GLUCOSE — a glucose load can precipitate it	Largely supportive; prevention is the treatment

The thiamine-before-glucose rule is examined, and you will one day need it on a real ward.

Managing alcohol withdrawal

Benzodiazepines — gold standard

- Long-acting: diazepam, chlordiazepoxide
- Lorazepam if liver disease
- Symptom-triggered against the CIWA-Ar scale, or fixed-schedule dosing

Supportive

- Thiamine, fluids, electrolytes (Mg, K, glucose)
- Anticonvulsant cover if seizures
- Monitor and treat in the right setting

Complications of chronic alcohol use

Alcoholics who keep drinking have a shortened life-span of roughly 15 years — through damage across nearly every organ system.

Hepatic & GI

Fatty liver → hepatitis → cirrhosis. Gastritis, pancreatitis, oesophageal varices. GI and liver cancers.

Neuro & cardiac

Peripheral neuropathy, cerebellar degeneration, dementia, Wernicke–Korsakoff. Cardiomyopathy, arrhythmia, hypertension.

Other systems

Macrocytosis, thrombocytopenia. Sexual dysfunction, hypogonadism. Fetal alcohol syndrome.

Medications for alcohol use disorder

Drug	Mechanism	Notes
Naltrexone (first-line)	μ-opioid antagonist → reduced craving and reward	Avoid in liver failure or current opioid use
Acamprosate (first-line)	Glutamate / GABA modulation	Maintains abstinence; renal dosing
Disulfiram (second-line)	Aldehyde dehydrogenase inhibitor → aversive reaction (flushing, nausea, tachycardia)	Adherence-dependent; no longer first-line

Plus psychosocial care: AA, motivational enhancement, CBT, relapse prevention. Do not confuse NALTREXONE (alcohol anti-craving) with NALOXONE (opioid-overdose reversal) — a classic exam and prescribing error.

What is he dependent on?

A 41-year-old businessman comes to the ED with insomnia for 3 days after running short of his "sleeping pills".

He asks specifically for a drug made by Roche, knows each tablet is 2 mg, and says he takes 5 tablets each night to sleep.

Most likely problem?

A Heroin abuse

B Benzodiazepine abuse

C Methadone abuse

D Painkiller abuse

Commit to an answer before the next slide.

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B — benzodiazepine abuse.

A 2 mg branded hypnotic taken in escalating doses is a benzodiazepine. Tolerance and dependence have developed; abrupt cessation causes rebound insomnia, anxiety and — dangerously — seizures.

Benzodiazepines & related sedatives

Pharmacology

- Potentiate GABA at GABA-A (chloride channel)
- Cross-tolerance and cross-dependence with alcohol
- Intoxication and withdrawal resemble alcohol

Intoxication — DSM-5

- Slurred speech; incoordination; unsteady gait
- Nystagmus
- Cognitive impairment (attention, memory)
- Stupor or coma

Overdose & withdrawal

- Flumazenil is the antagonist for overdose — may precipitate seizures
- Withdrawal is like alcohol — CAN BE FATAL
- Gradual taper, never abrupt stop

Opioids — types & intoxication (DSM-5)

Types

- Natural: opium, morphine, codeine
- Semi-synthetic: heroin
- Synthetic: fentanyl, pethidine, methadone

Intoxication criteria

- A. Recent opioid use
- B. Problematic behavioural change — euphoria then apathy, dysphoria, psychomotor change, impaired judgment
- C. Pupillary constriction (or dilation from anoxia in severe overdose) plus ≥ 1 of drowsiness/coma, slurred speech, attention or memory impairment

The killer & the antidote

- Pinpoint pupils, bradycardia, constipation
- RESPIRATORY DEPRESSION is the cause of death
- Overdose → naloxone plus airway and oxygen
- Take-home naloxone kits save lives

What is happening to this prisoner?

A 33-year-old man is jailed after reckless high-speed driving at 3 a.m.

The next day he develops excessive lacrimation, a runny nose, repeated vomiting and abdominal cramps. His consciousness is fully intact.

Most likely problem?

A Cannabis abuse

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C Naloxone abuse

D Opioid withdrawal

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D **Opioid withdrawal**

D — opioid withdrawal.

Lacrimation, rhinorrhoea, vomiting, cramps and dilated pupils with intact consciousness = opioid withdrawal, precipitated here by incarceration and cessation. Miserable, but not life-threatening.

Opioid withdrawal — DSM-5 criteria

A · B (≥ 3, minutes to days)

- Dysphoric mood
- Nausea or vomiting
- Muscle aches
- Lacrimation or rhinorrhoea
- Pupillary dilation, piloerection, or sweating
- Diarrhoea; yawning; fever; insomnia

C · D · course

- Craving from ~6 h, peaks at 36–48 h
- Not life-threatening, but very distressing
- Graded with COWS — pulse, pupils, tremor

Key contrast: alcohol and sedative withdrawal can kill; opioid withdrawal makes patients feel awful but is rarely fatal.

Opioid treatment & medication-assisted treatment

Symptom control

Clonidine or lofexidine (α2-agonists; lofexidine FDA-approved 2018) for autonomic symptoms. NSAIDs for pain, loperamide for diarrhoea, antiemetics.

Maintenance (MAT)

Methadone — full μ-agonist, clinic-based. Buprenorphine/naloxone — partial agonist, office-based. Naltrexone XR — antagonist, after detox.

Harm reduction

Needle and syringe exchange. Take-home naloxone kits. Blood-borne virus screening.

Complications of injecting drug use

Local

- Track marks
- Cellulitis, abscess
- Thrombophlebitis

Systemic

- Infective endocarditis
- Sepsis
- Vascular damage, DVT

Blood-borne

- HIV
- Hepatitis B and C

Track marks and early cellulitis with multiple recent injections point to heroin use — examine the injection sites and screen for blood-borne viruses in every injector.

What has this teenager used?

A 16-year-old boy presents with slurred speech, incoordination and nausea.

Examination reveals a rash on the skin around his mouth and nose. When asked about substance use, he answers yes.

Most likely substance?

A Cannabis

B Alcohol

C Volatile substance (inhalant)

D Morphine

Commit to an answer before the next slide.

What has this teenager used?

ANSWER

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C **Volatile substance (inhalant)**

D Morphine

C — a volatile substance (inhalant).

CNS depression in an adolescent with a perioral rash — "glue-sniffer's rash" — is classic inhalant use. Beware "sudden sniffing death" from arrhythmia.

Inhalants (volatile substances)

What & who

- Volatile organics: glue, solvents, acetone, benzene, aerosols, nitrous oxide
- CNS depressants; adolescent experimentation
- Rapid, brief effect

Presentation & harm

- Intoxication like other depressants; perioral rash
- Multi-organ damage — brain, liver, kidney, marrow
- "Sudden sniffing death" — fatal arrhythmia, even on first use

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PAUSE

Break — 10 minutes

After the break: stimulants and the regional picture — Captagon and khat · hallucinogens · cannabis and the endocannabinoid system · novel substances · then the pattern-recognition tables and the case.

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PART 3

Stimulants, Hallucinogens & Cannabis

The substances that activate — plus hallucinogens, cannabis and novel drugs, with the Saudi and regional picture.

What is driving his behaviour?

Khalid, a 20-year-old man, is brought to the ER by police who arrested him for reckless driving (high-speed drifting) and violent behaviour.

He looks over-suspicious, agitated, and over-talkative.

Most likely?

A Opioid intoxication

B Stimulant intoxication

C Alcohol intoxication

D Cannabis withdrawal

Commit to an answer before the next slide.

What is driving his behaviour?

ANSWER

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He looks over-suspicious, agitated, and over-talkative.

Most likely?

A Opioid intoxication

B **Stimulant intoxication**

C Alcohol intoxication

D Cannabis withdrawal

B — stimulant intoxication.

Agitation, paranoia, hyperactivity and pressured over-talkativeness point to a stimulant. In the KSA and regional context, think amphetamine / Captagon.

Stimulants — overview & clinical effects

The drugs

- Cocaine
- Amphetamine / methamphetamine
- MDMA
- Captagon, khat
- Caffeine, nicotine

Psychological effects

- Elevated mood, euphoria
- Increased confidence, self-esteem, sociability; over-talkativeness
- Overactivity, insomnia
- High dose: paranoid psychosis, visual hallucinations, aggression, hostility

Physical effects

- Reduced fatigue and appetite
- Dilated pupils, tremor, raised HR / BP / temperature
- High dose: arrhythmia, severe hypertension, hyperthermia, seizures, stroke

Cocaine

Forms & routes

- Cocaine HCl — intranasal or intravenous
- Freebase / crack — smoked
- Rapid, intense, short-lived high → bingeing

Harms

- Cardiac: arrhythmia, myocardial infarction, aortic dissection
- Neuro: seizures, stroke
- Nasal septum perforation with intranasal use

Captagon & khat — the regional picture

Amphetamine-type stimulants are the leading substance among patients in Saudi addiction treatment — reported in roughly 40–70% depending on centre and era, and in 48% of 12,743 first admissions to Al-Amal Dammam over 1986–2006.

Captagon (fenethylline)

- An amphetamine-type stimulant in tablet form
- THE dominant illicit stimulant in Saudi Arabia and the region
- Paranoia, aggression, insomnia, cardiovascular risk

Khat (Catha edulis)

- Leaves chewed; the active alkaloid is cathinone
- Common in south-west KSA and Yemen
- Mild to moderate stimulant
- Oral and dental disease, psychosis, dependence



Catha edulis (khat). User:Katpatuka, Wikimedia Commons, public domain

AbuMadini et al., Drug Alcohol Depend 2008. This is the regionally essential slide — in this setting, think Captagon before cocaine.

Stimulant intoxication — DSM-5 criteria

A · B · D

- A. Recent use of an amphetamine-type substance, cocaine, or other stimulant
- B. Problematic behavioural or psychological change — euphoria or affective blunting; changed sociability; hypervigilance; interpersonal sensitivity; anxiety, tension or anger; stereotyped behaviour; impaired judgment
- D. Not due to another condition

C · ≥ 2 signs

- Tachycardia or bradycardia
- Pupillary dilation
- Elevated or lowered blood pressure
- Perspiration or chills; nausea or vomiting
- Weight loss; psychomotor agitation or retardation
- Weakness, respiratory depression, chest pain, arrhythmia
- Confusion, seizures, dyskinesias, dystonias, or coma

Stimulant withdrawal — the "crash"

A · B · C · D

- A. Cessation or reduction of prolonged stimulant use
- B. Dysphoric mood plus ≥ 2 physiological changes within hours to days: fatigue; vivid unpleasant dreams; insomnia or hypersomnia; increased appetite; psychomotor retardation or agitation
- C. Significant distress or impairment
- D. Not due to another condition

Clinical

- Abrupt abstinence is NOT life-threatening
- Post-intoxication depression — the "crash": malaise, hypersomnolence, anhedonia, hunger
- CAN BECOME SUICIDAL — monitor mood and safety

Treating stimulant intoxication & withdrawal

Intoxication / agitation

- Mild: reassurance plus benzodiazepines
- Severe agitation or psychosis: an antipsychotic such as haloperidol
- Treat hyperthermia aggressively with cooling; control BP and arrhythmia

Withdrawal

- Supportive; monitor mood and suicide risk
- NO approved medication-assisted treatment
- Psychosocial care — CBT, contingency management

Jealousy, insomnia and hypervigilance

A 32-year-old man has become increasingly irritable, insomniac and hypervigilant over the past 4 weeks, with an unpredictable mood.

He accuses his wife of extramarital relationships.

Most likely diagnosis?

A Heroin abuse

B Generalized anxiety disorder

C Amphetamine abuse

D Paranoid schizophrenia

Commit to an answer before the next slide.

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C — amphetamine abuse.

Stimulant-induced paranoia and morbid jealousy with hypervigilance and insomnia. Screen for stimulant use before diagnosing a primary psychotic disorder.

Caffeine

Intoxication

- The world's most widely used psychoactive drug
- Restlessness, insomnia, diuresis, GI upset
- Muscle twitching, tachycardia or arrhythmia at high doses

Withdrawal

- Headache, lethargy, low mood
- Difficulty concentrating, irritability
- Onset ~12–24 h; lasts 2–9 days

DSM-5 recognises caffeine intoxication and withdrawal as disorders — "legal" and "everyday" do not mean "without a withdrawal syndrome".

Hallucinogens (psychedelics)

Drugs & effects

- LSD (synthetic), psilocybin (magic mushrooms)
- Perceptual distortion; visual and tactile hallucinations
- Altered sense of time and space; euphoria, anxiety, panic, paranoia
- Flashbacks (HPPD); physically: mydriasis, raised HR / BP / temperature

Dependence

- Minimal tolerance
- NO physical dependence and no withdrawal syndrome
- The risk is from behaviour under the influence — accidents, trauma, "bad trips", occasionally suicide or violence

Red eyes and disinhibition

A 43-year-old man has episodic behavioural disturbances including euphoria, talkativeness and disinhibition.

His eyes look red most of the time.

Most likely diagnosis?

A Alcohol abuse

B Cannabis abuse

C Amphetamine abuse

D Cocaine abuse

Commit to an answer before the next slide.

Red eyes and disinhibition

ANSWER

A 43-year-old man has episodic behavioural disturbances including euphoria, talkativeness and disinhibition.

His eyes look red most of the time.

Most likely diagnosis?

A Alcohol abuse

B **Cannabis abuse**

C Amphetamine abuse

D Cocaine abuse

B — cannabis abuse.

Conjunctival injection ("red eyes") with euphoria, disinhibition and increased appetite is the cannabis intoxication picture.

Cannabis intoxication — DSM-5 criteria

The active ingredient is THC, a partial CB1 agonist.

A · B · D

- A. Recent use of cannabis
- B. Problematic behavioural or psychological change — impaired motor coordination, euphoria, anxiety, sensation of slowed time, impaired judgment, social withdrawal
- D. Not due to another condition

C · ≥ 2 within 2 hours

- Conjunctival injection (red eyes)
- Increased appetite
- Dry mouth
- Tachycardia



Cannabis sativa leaf. Wikimedia Commons, public domain

Cannabis withdrawal — DSM-5 criteria

A · B (≥ 3 within ~1 week)

- Irritability, anger or aggression
- Nervousness or anxiety
- Sleep difficulty — insomnia, disturbing dreams
- Decreased appetite or weight loss
- Restlessness; depressed mood

Plus a physical symptom · C · D

- ≥ 1 physical symptom causing discomfort: abdominal pain, tremor or shakiness, sweating, fever, chills, or headache
- Occurs in HEAVY, DAILY users
- Not dangerous — but a major reason people relapse

Recognising and normalising cannabis withdrawal supports abstinence — patients often assume it means something is wrong with them.

Cannabis products & CBD

Forms — rising potency

- Marijuana — dried leaves and flowers
- Hashish — compressed resin
- Hash oil — concentrated extract, the highest THC

THC vs CBD

- THC — the psychoactive, intoxicating component
- CBD (cannabidiol) — non-intoxicating; studied for epilepsy (Epidiolex) and anxiety

Potency has risen sharply over recent decades — which matters clinically because psychosis risk is dose- and potency-dependent.

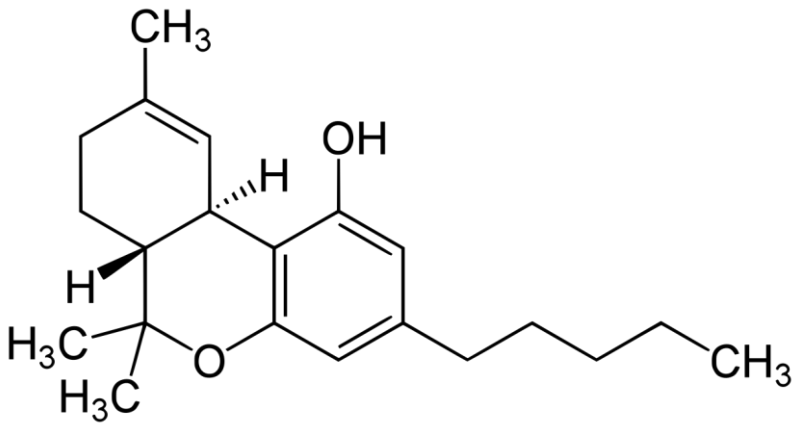
The endocannabinoid system

Components

- Receptors: CB1 (central nervous system) and CB2 (immune system)
- Endogenous ligands: anandamide and 2-AG

Roles & THC

- Regulates appetite, mood, memory, pain, immunity, neurodevelopment
- THC acts on CB1 → the psychoactive effects
- Because CB1 shapes neurodevelopment, heavy ADOLESCENT use is particularly concerning



Δ9-tetrahydrocannabinol. Yikrazuul, Wikimedia Commons, public domain

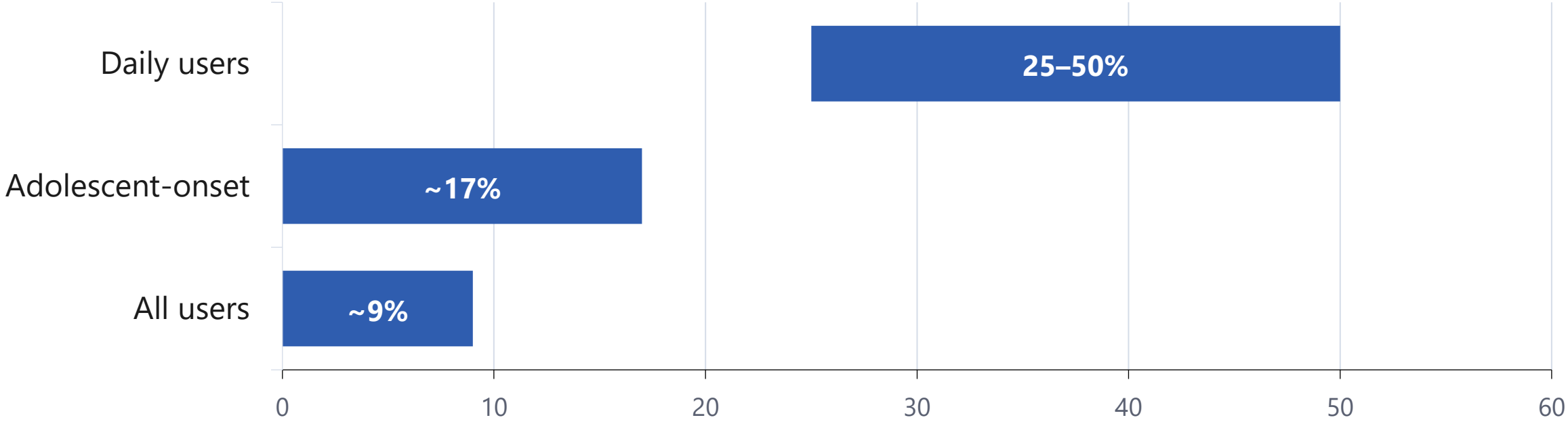
Dependence, psychosis & the Saudi picture

~9%
of all users develop a cannabis use disorder

~17%
if use begins in adolescence

25–50%
among daily users

Risk of developing a cannabis use disorder (%), by pattern of use



Adolescent, frequent, high-potency use raises psychosis and schizophrenia risk in a dose-dependent way. KSA (MOH Al-Amal): cannabis and amphetamines are the leading substances presenting for treatment.

Novel psychoactive substances (NPS)

The families

- Synthetic cannabinoids — "Spice", "K2"
- Synthetic cathinones — "bath salts"
- Synthetic opioids — fentanyl analogues
- Designer benzodiazepines

Why they're dangerous

- Highly variable potency — some fentanyl analogues up to ~10,000× morphine
- Unpredictable, severe toxicity
- Often MISSED on standard drug screens

Suspect NPS when the clinical picture is extreme but the routine screen is negative.

Nicotine — pharmacokinetics & treatment

Pharmacokinetics

- Reaches the brain in 7–10 SECONDS
- Withdrawal begins ~30 minutes after the last cigarette
- Cotinine detectable 7–14 days
- Faster metabolisers → higher dependence and cancer risk

Treatment

- Varenicline — α4β2 nicotinic partial agonist
- Bupropion — dopamine/noradrenaline reuptake inhibitor
- NRT — patch, gum, lozenge, spray, inhaler
- ALWAYS add behavioural support

Vaping / e-cigarettes deliver nicotine as an aerosol, often at high dose (one pod ≈ a pack of cigarettes) — a growing driver of adolescent dependence. Not harmless: EVALI is linked to vitamin-E acetate in THC vapes.

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PART 4

Integration & Review

Pattern-recognition tables · the withdrawal-danger rule · board questions · and sources.

Intoxication — at a glance

Reach for the pupils first: PINPOINT means opioids · DILATED means stimulants or hallucinogens.

Substance	Action	Pupils	Key signs & symptoms
Cocaine / amphetamine	Stimulant	Dilated	Raised HR/BP/temp/energy, paranoia, euphoria, reduced fatigue and appetite, impaired judgment, hallucinations, prolonged wakefulness
Alcohol / sedatives	Sedative	Normal	Slurred speech, ataxia, disinhibition, reduced respiration → coma and death
Heroin / opioids	Sedative, analgesic	Pinpoint	Drowsiness and "nodding", euphoria, constipation, reduced respiration, seizures; look for track marks
Cannabis	Euphoriant	Normal	Red eyes, increased appetite, slowed time, reduced intraocular pressure; high dose → hallucinations
Caffeine	Stimulant	Normal	Restlessness, insomnia, diuresis, muscle twitching, arrhythmia
Nicotine	Stimulant	Normal	Mild arousal; most symptoms come from withdrawal

Withdrawal — and which ones can kill

Substance	Onset	Duration & key signs	Danger
Alcohol	6–72 h	5–7 days; tremor, autonomic storm, seizures, DTs	DANGEROUS
Sedatives / benzodiazepines	Variable (half-life)	Similar to alcohol; seizures	DANGEROUS
Opioids	6–48 h	4–7 days; lacrimation, GI upset, aches, dilated pupils	Not life-threatening
Stimulants	Hours–days	3–4 day "crash"; fatigue, hypersomnia, low mood	Watch suicidality
Cannabis	~1 day	Up to weeks; irritability, insomnia, reduced appetite	Mild
Caffeine / nicotine	2–24 h	Days–weeks; headache, irritability, craving	Not dangerous

ONLY alcohol and sedative/benzodiazepine withdrawal are directly life-threatening (seizures, DTs). Opioid and stimulant withdrawal are miserable but rarely fatal.

Antidotes & overdose — quick reference

Emergency	First step / antidote
Opioid overdose (pinpoint pupils, low RR)	Naloxone + airway and oxygen — short-acting, may need repeating; can precipitate withdrawal
Sedative / benzodiazepine overdose	Supportive airway and ventilation; flumazenil rarely used — it can precipitate seizures, especially in dependence or mixed overdose
Alcohol withdrawal / DTs	Benzodiazepines + thiamine + fluids and electrolytes
Wernicke's encephalopathy	THIAMINE BEFORE GLUCOSE
Stimulant toxicity	No antidote — benzodiazepines, active cooling, control BP and arrhythmia

Two antidotes to know cold: naloxone for opioids and thiamine for Wernicke's. Flumazenil is usually avoided — the seizure risk outweighs the benefit.

Mr. K, 48 — day 3 after a hip repair

Mr. K, 48, was admitted after a fall and had a hip repair two days ago. The surgical team calls you because he has become "agitated and confused overnight".

He is tremulous and drenched in sweat, pulse 118, blood pressure 168/98, temperature 37.8 °C. He is disoriented to place and time, pulling at his lines, and tells you there are insects crawling on the bed sheets. He was lucid and pleasant on the ward round yesterday morning.

His wife, reached by phone, says he drinks "most evenings" — she thinks around a bottle of spirits over two or three days — and that he has had "shaking in the mornings" for years, which settles after his first drink. He has never been admitted for this.

Nursing notes record that he had a brief seizure about six hours ago. His MCV is raised and GGT is markedly elevated; his nutrition on the ward has been poor.

Case problem — work through it

Questions

- 1 · What is the diagnosis, and what in the timeline confirms it?
- 2 · Why is this an emergency — what is the untreated mortality?
- 3 · What is your first-line treatment, and what must you avoid?
- 4 · What must you give before any glucose, and why?
- 5 · What will you offer him for the longer term?

Model answer

- 1 · Delirium tremens — onset 48–72 h after the last drink (lucid on day 2), preceded by a withdrawal seizure at 6–48 h
- 2 · DTs is the one withdrawal syndrome that kills — from arrhythmia, seizures, hyperthermia and collapse. It needs monitoring, not a quiet side room
- 3 · Benzodiazepines, symptom-triggered against a withdrawal scale, plus fluids, electrolytes and a search for infection. AVOID an antipsychotic as first-line — it lowers the seizure threshold
- 4 · Thiamine before glucose — carbohydrate in a thiamine-depleted patient can precipitate Wernicke's, which untreated becomes irreversible Korsakoff amnesia
- 5 · After detox: naltrexone, acamprosate or disulfiram; psychosocial treatment; screening for dual diagnosis and chronic alcohol-related harm

Finish by asking what would change if the same picture appeared on day 1 rather than day 3 — the answer being simple withdrawal or another cause of delirium entirely.

Board-style questions

#	Question	Answer
1	Which two withdrawal syndromes are directly life-threatening?	Alcohol and sedative/benzodiazepine withdrawal — seizures, DTs
2	Pinpoint pupils plus respiratory depression — diagnosis and antidote?	Opioid intoxication → naloxone, plus airway and oxygen
3	First-line medications for alcohol use disorder?	Naltrexone and acamprosate (disulfiram is second-line)
4	Confusion + ophthalmoplegia + ataxia — diagnosis and treatment?	Wernicke's encephalopathy → thiamine BEFORE glucose
5	Adolescent with slurred speech, ataxia and a perioral rash?	Inhalant (volatile substance) use

If you can answer these five, you can navigate most SUD stations.

Summary — six things to remember

Golden rules

- Addiction is a chronic, relapsing brain disease (dopamine reward); relapse ≠ failure
- Diagnose with DSM-5 (≥ 2 of 11) or the 4 C's; grade severity
- Only alcohol and sedative withdrawal can kill — treat with benzodiazepines

…continued

- Opioid overdose = pinpoint pupils + low RR → naloxone; MAT = buprenorphine or methadone
- Alcohol MAT = naltrexone / acamprosate; nicotine = varenicline / NRT
- Screen for comorbidity; psychosocial care and faith/community support are central

References

Key figures were cross-checked against these primary sources.

Classification & guidelines

- APA — DSM-5-TR (2022)
- NICE and ASAM guidelines on substance misuse
- SAMHSA treatment guidance

Evidence & regulatory

- Kaplan & Sadock's Synopsis of Psychiatry
- WHO global status report on alcohol and health and treatment of substance use disorders (2024)
- UNODC World Drug Report 2026 — 2024 reference year
- FDA — lofexidine (2018), naltrexone, varenicline
- AbuMadini et al., Drug Alcohol Depend 2008 — Al-Amal Dammam, n = 12,743
- Ibrahim et al., Ann Saudi Med 2018;38(5):319–325 — Al-Qassim, n = 612

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Thank You

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Questions welcome. Sources for every figure are on the references slide.